

ASSESSMENT 2 REPORT: AUTOMATED POTHOLE DETECTION AND SEVERITY GRADING USING TRANSFER LEARNING

**MCTA 4363
DEEP LEARNING**

SECTION 1

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GROUP NAME: MIXUE

No.	Name	Matric No.
1	Noramira Husna Binti Norkhairani	2214496
2	Nur Amira Nazira Binti Mohd Nasir	2110026
3	Ahmad Adam Bin Norhelme	2219973

1. INTRODUCTION

Potholes represent a critical global road infrastructure challenge, posing significant risks to public safety and often causing major traffic incidents. Municipalities worldwide struggle with resource constraints, often limited to reactive repairs rather than proactive maintenance. Consequently, road users must actively navigate these hazards while seeking to avoid vehicle damage and potential accidents.

Artificial intelligence and deep learning technologies offer promising solutions for automating the assessment of pothole severity, enabling distinction between negligible flaws and high-risk hazards. This project investigates two distinct deep learning approaches for classifying pothole severity:

- Single-stage YOLOv8 Segmentation Model - An instance segmentation architecture that generates precise pixel-level masks for potholes
- Two-stage Heuristic Convolutional Neural Network (CNN) - A detection-based approach that isolates potholes and applies heuristic formulas for severity classification

The primary objective is to evaluate and determine which architecture provides superior accuracy and efficiency for real-world severity classification, ultimately contributing to automated driver assistance systems and infrastructure monitoring.

2. PROBLEM STATEMENT

Traditional road monitoring relies on manual inspections or vibration sensors, which are either slow, labor-intensive, or prone to false alarms. While vision-based deep learning has become the standard approach for pothole detection, several significant challenges remain:

Key Challenges:

- Severity Grading Difficulty: Detecting a pothole is relatively straightforward, but grading its physical severity from standard 2D camera imagery remains a major challenge due to the lack of depth information.
- Cost Constraints: Researchers often rely on expensive 3D sensors like LiDAR to measure volume, which is not scalable for widespread deployment.
- Monocular Domain Gap: Objects appear smaller as they approach the vanishing point in camera frames, making accurate size estimation from 2D images problematic.
- Dataset Complexity: Dashcam-perspective imagery includes varying lighting conditions, shadows, and road textures, creating a rigorous benchmark for classification systems.
- Limited Generalization: Models trained on specific datasets may fail to perform on different road conditions, camera angles, or distances.

This project addresses these challenges by developing and comparing two automated approaches that can grade pothole severity using only monocular vision, providing a more affordable and scalable solution for infrastructure monitoring.

3. METHODOLOGY

3.1 Dataset and Ground Truth Establishment

The proposed dataset consists of dashcam-perspective imagery specifically curated to represent real-world driving conditions. The dataset includes:

- Varying lighting conditions
- Shadows
- Different road textures

Ground Truth Labeling: Each pothole was manually annotated by establishing bounding boxes and assigning severity classifications based strictly on spatial dimensions and visual footprint. The classification follows the ASTM D6433-16 standard for Pavement Condition Index (PCI) surveys, which categorizes pothole severity into three classes:

Severity Level Criteria

Low	Depth: 13-25mm, Diameter: 100-200mm
Medium	Depth: 25-50mm, Diameter: 200-450mm
High	Depth: >50mm, Diameter: >450mm

A total of 296 images with labels 0, 1, and 2 (representing minor, moderate, and severe potholes respectively) were prepared for the classification dataset.

3.2 Severity Estimation Heuristics

Across both methodologies, the final severity classification relies on a shared heuristic algorithm that maps 2D pixel data to physical threat levels. Once a Region of Interest (ROI) is established, the system extracts a Size Proxy Score (S_{size}) to determine the hazard level.

Perspective-Aware Scaling: To account for the monocular domain gap, the raw pixel area of the detected hazard is dynamically scaled based on its vertical spatial coordinate (y-axis) within the frame. The proximity of the bounding box or mask to the bottom of the frame (the camera's immediate foreground) dictates the scaling factor applied to the raw area. This

normalized S_size is then evaluated through a discrete classification threshold matrix to assign the final severity of Low, Medium, or High.

3.3 Benchmarked Architectures

Method 1: Two-Stage Heuristic Convolutional Neural Network (CNN)

This approach optimizes processing speed through a two-stage process:

- A CNN acts as a detector to isolate potholes in the image
- Once localized, severity is calculated using a heuristic formula based on bounding box dimensions relative to their position in the camera frame
- This method bypasses the need for complex segmentation processing, potentially offering faster inference times.

Method 2: YOLOv8 Instance Segmentation

This method employs YOLOv8 to generate precise pixel-level masks for each pothole:

- The model performs instance segmentation, creating accurate boundaries for each detected pothole
- Because the mask inherently excludes background asphalt, severity classification is derived directly from the exact pixel footprint of the mask

This approach provides the most accurate spatial measurement as it eliminates noise found in bounding-box approximations.

3.4 Evaluation Metrics

The models were evaluated using standard object detection and segmentation metrics:

- Precision: Ratio of correctly detected potholes to total detections
- Recall: Ratio of correctly detected potholes to total ground truth instances
- mAP@0.5: Mean Average Precision at IoU threshold 0.5
- mAP@0.5:0.95: Mean Average Precision averaged over IoU thresholds from 0.5 to 0.95

4. RESULTS AND DISCUSSION

4.1 Performance Metrics

The experimental analysis compared the performance of the Two-Stage Heuristic CNN and the YOLOv8 Instance Segmentation model. The comparative metrics are summarized in Table 1.

Table 1: Comparative Performance Metrics

Metric	Instance Segmentation	2-stage Heuristic CNN
Precision	0.308	0.197
Recall	0.105	0.110
mAP@0.5	0.0726	0.0559
mAP@0.5:0.95	0.0726	0.0159

4.2 Analysis of Results

4.2.1 Precision and Localization

The Instance Segmentation model achieved significantly higher precision (0.308) compared to the Heuristic CNN (0.197). This confirms that pixel-level mask generation is more effective at filtering out non-hazardous road features than standard bounding-box approximations.

The most significant finding lies in the mAP@0.5:0.95 metric, where Instance Segmentation (0.0726) outperformed the Heuristic CNN (0.0159) by a factor of nearly five. This indicates that segmentation masks provide a much higher degree of localization accuracy, ensuring that hazard boundaries are more precisely identified.

4.2.2 Severity Classification Behavior

The confusion matrix analysis revealed that the model exhibits a bias toward the "minor" severity category, indicating that the heuristic logic is currently calibrated to be conservative. This suggests that the current classification threshold may be overly sensitive to minor surface variations, requiring further optimization in the heuristic grading logic.

4.2.3 Model Performance Challenges

The lower metrics across both models are indicative of the extreme difficulty of the dashcam-perspective dataset, which provides a rigorous benchmark for future hazard classification systems. The performance parity in Recall suggests a "precision-recall trade-off": while the CNN maintains a slightly higher recall (0.110), the Instance Segmentation model's superior precision and higher mAP scores demonstrate that it is the more stable architecture for autonomous road maintenance.

4.3 Key Findings

Superior Architecture: Instance-based feature extraction (YOLOv8 Segmentation) is the optimal path for high-accuracy road hazard assessment

Localization Advantage: Pixel-level masks provide significantly better spatial accuracy compared to bounding-box approximations

Rule-Based Limitation: The severity classification pipeline is rule-based and not learned within the model, which limits its ability to adapt to complex real-world scenarios

Validation Issues: The model was not able to validate drone-style imagery data effectively, indicating generalization challenges

5. CONCLUSION

Although the original objectives of achieving high classification accuracy were not fully met, the project successfully accomplished several key deliverables:

- A working video segmentation pipeline with severity overlay functionality
- A pothole severity classification dataset with 296 images labeled with severity classes (0, 1, 2 representing minor, moderate, and severe potholes)

The experimental results demonstrate that the Instance Segmentation architecture provides measurable advantages in spatial reliability and localization accuracy compared to the heuristic CNN approach. However, both models face challenges with complex dashcam-perspective datasets.

5.1 Future Improvements

Several enhancements are recommended for future work:

- Generalized Dataset Training: Fine-tune the model on a more generalized dataset to consider different road conditions, camera angles, and distances from the road. This aims to improve model robustness and accuracy.
- Neural Network Classifier: Replace the heuristic severity classifier with a neural network classifier that takes the masked region as input. This would allow the model to focus more on depth and texture features, potentially improving severity classification accuracy.
- Extended Training: Train the model with more epochs and increased variability to further generalize the model's performance across diverse scenarios.
- Multi-Sensor Integration: Consider incorporating additional data sources (such as depth sensors) to enhance the monocular vision approach.

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